

Comparative Analysis of Reconfigurable Intelligent Surfaces and Fluid Antenna System Relays for 6G

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Abstract—The evolution to 6G wireless networks requires breakthrough technologies that drastically improve energy and spectral efficiency. To study the energy and spectral efficiency of emerging relaying systems, this paper presents a comparative analysis of active Reconfigurable Intelligent Surfaces (RIS), passive RIS, and Fluid Antenna System (FAS) relay technologies under actual deployment conditions. Spatially coupled channel models, power consumption profiles, and performance optimization algorithms are investigated to enable fair comparison. The data rate and energy efficiency are examined using Monte Carlo simulations under various signal-to-noise ratio (SNR) and array size. Our findings reveals that no single technology dominates all operating scenarios. Active RIS performs best at low SNR, while FAS relay with Decode-and-Forward protocol (DF) excels high SNR. Passive RIS suits small deployment distance with low energy requirement. These results are vital to identify for developing flexible strategies for deploying 6G relaying networks that choose suitable technologies based on operational needs.

Index Terms—Fluid antenna system (FAS), reconfigurable intelligent surfaces (RIS), relay, decode-and-forward protocol (DF).

I. INTRODUCTION

The rapid increase in wireless data traffic and the advent of new applications demanding ultra-reliable, low-latency communications have compelled researchers to investigate advanced technologies for future 6G wireless networks [1]. Traditional communication frameworks face significant challenges in meeting these high demands, particularly in bad channel conditions. Reconfigurable Intelligent Surfaces (RIS) represent a revolutionary technology that contests the long-standing belief that the wireless channel is an uncontrolled phenomenon that must just be accommodated. RIS technology facilitates the exact modification of incident wavefronts in the propagation environment by integrating programmable electromagnetic structures, allowing for software-controlled reflection, refraction, absorption, or polarization [2]. This exceptional control over the wireless medium is expected to improve signal quality. RIS technology divided into two main architectural paradigms: passive RIS, which functions without active power-consuming components and depends on intelligent phase shifting of incident signals; and active RIS, which includes low-noise amplification to mitigate the double-fading phenomenon associated with passive RIS implementations [3]. Recent large-scale studies show that both methods have certain benefits and drawbacks that call for close consideration in diverse deployment settings [4].

The low power consumption and easy hardware design of passive RIS designs have drawn evolution attention. [5] examined passive RIS phase optimization approaches and showed that correctly constructed passive surfaces may enhance channel capacity without active amplification. [6] proposed created analytical models that show the basic trade-offs between performance gains and hardware complexity. It was shown that passive RIS systems are more energy efficient than traditional relay systems when the number of reflecting elements goes over a certain threshold. However, it was found that "double-fading" was a significant problem. This is when signals attenuate as they go through both the transmitter-to-RIS and RIS-to-receiver channels. Active RIS use amplification to overcome the limitations of passive systems, notably the double-fading problem. [7] defined active RIS designs by amplification techniques, power consumption profiles, and performance parameters. [3] found that active RIS designs outperformed passive metal plate reflectors by 10 dB, overcoming passive RIS's "double fading" constraint. Since energy consumption is essential for active RIS implementation.

On the other hand, fluid antenna systems which uses software-controllable fluid medium to dynamically reconfigure its shape, size, position, length, direction, and radiation characteristics are capable to achieve extremely high spatial diversity with minimal hardware complexity. FAS relay optimize antenna position and selectively activate ports at multiple preset locations within a specified area in real time, improving energy efficiency if compared to traditional relay with fixed-position antenna systems [8]. A comparative study quantifying their relative advantages if compared to RIS based system in the same operational contexts is notably lacking in the literature. The research gap is especially evident in energy efficiency, a critical parameter in forthcoming 6G systems [9].

This paper presents a comparative analysis of active RIS, passive RIS, and FAS relay technologies in terms of achievable data rate and energy efficiency. The main contributions are:

- A uniform analytical framework is developed for active RIS, passive RIS, and FAS relay systems with spatially correlated realistic channel models and energy model;
- Non-monotonic relationship between system performance and operational parameters is revealed through extensive Monte Carlo simulations.
- The scaling behavior of each technique is analyzed with increasing elements/ports.

The subsequent sections of the paper are structured as follows: Section II outlines the system model and formulates the problem. Section III presents the optimization formulation. Section IV presents the numerical simulation results while. Section V concludes the paper.

II. SYSTEM MODEL

We examine a three-node wireless communication system with a base station (BS), user equipment (UE), and intermediate relay node. FAS relay and RIS are considered as intermediary relay nodes. A square array of dimension $l \times l$ is considered

For fairness, in our comparison framework, all technologies use equal physical panel dimensions while meeting their inter-element spacing needs, resulting in technology-specific element densities:

- RIS elements: $N_{RIS} = [W/(\lambda/2)]^2$, where $\lambda/2$ spacing avoids grating lobes
- FAS ports: $N_{FAS} = [W/(\lambda/4)]^2$, where $\lambda/4$ spacing enables tighter packing

where $\lambda = c/f_c$ is the wavelength and $c = 3 \times 10^8$ m/s is the speed of light. W is the physical side length (in meters) of the square antenna array panel that constrains both RIS and FAS technologies to the same deployment area. This results in $N_{FAS}/N_{RIS} \approx 4$, maintaining the physical size ratio between technologies while respecting their fundamental design principles. To avoid spatial aliasing and grating lobes in the radiation pattern and optimize beamforming, the half-wavelength spacing criteria is essential. FAS's quarter-wavelength spacing improves spatial variety exploitation through selective port activation while increasing port density and resolution.

A. Channel Model

The wireless channel is assumed to experience Rician fading and path loss. The channel coefficient from the BS to the i th port or element is

$$h_{BR,i} = \sqrt{PL_{BR}} \left(\sqrt{\frac{K}{K+1}} h_{LOS,i} + \sqrt{\frac{1}{K+1}} h_{NLOS,i} \right) \quad (1)$$

where PL_{BR} denotes the path loss, K is the Rician factor quantifying the power ratio between line-of-sight (LOS) and non-line-of-sight (NLOS) components; $h_{LOS,i}$ is the LOS component, and $h_{NLOS,i} \sim CN(0,1)$ is the NLOS component. Similarly, $h_{RU,i}$ denotes the channel from the i th element/port to the UE.

Frequency-dependent path loss model accounts for both distance and frequency effects

$$PL(f, d) = PL_0 \left(\frac{f}{f_{ref}} \right)^2 \left(\frac{d}{d_0} \right)^\alpha L_{atm}(f) \quad (2)$$

where PL_0 is the reference path loss, f_{ref} is the reference frequency, α is the path loss exponent, and $L_{atm}(f) = 1 + 0.1 \left(\frac{f}{f_{ref}} - 1 \right)^{1.5}$ takes into account atmospheric absorption at high frequencies.

The spatial correlation between elements i and j is modeled using the exponential correlation model

$$A_{i,j} = E[h_i h_j] = \rho^{|i-j|} \exp(-j\psi_{i,j}) \quad (3)$$

where $\rho \in [0,1]$ is the correlation coefficient, $\psi_{i,j}$ is the phase difference, and $|i-j|$ represents the element separation. The correlation coefficient is

$$\rho = J_0(2\pi\Delta\cos(\theta)/\lambda) \quad (4)$$

where J_0 is the zero-order Bessel function of the first kind, and θ is the angle of arrival (AoA) or angle of departure (AoD) of the signal.

The complete spatial correlation matrix $A \in C^{(N \times N)}$ has entries

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1N} \\ A_{21} & A_{22} & \dots & A_{2N} \\ A_{31} & A_{32} & \dots & A_{3N} \\ \vdots & \vdots & \ddots & \vdots \\ A_{N1} & A_{N2} & \dots & A_{NN} \end{bmatrix} \quad (5)$$

Correlation characteristics of different array configurations:

- RIS array: $\lambda/2$ spacing results in a correlation coefficient of approximately 0.6-0.8 between adjacent elements.
- FAS array: $\lambda/4$ spacing results in higher spatial correlation, with a correlation coefficient of approximately 0.8-0.9 between adjacent ports.
- Large-scale array: As N increases, the correlation between edge elements approaches zero.

In our simulation, to balance computational complexity and modeling accuracy, we use a simplified exponential fading model $A_{ij} = \rho^{|i-j|}$, where ρ is the correlation coefficient parameter. We adopt $\rho = 0.7$ for RIS and $\rho = 0.8$ for FAS to provide conservative performance evaluation that accounts for realistic implementation conditions. This approach follows established practices in massive MIMO analysis where effective correlation models are used for system-level studies. High correlation restricts diversity increase but improves beamforming stability for Active RIS. Correlation limits the degrees of freedom for phase optimization in Passive RIS, therefore influencing signal merging efficiency. Regarding FAS relay, correlation affects the choice of port since linked ports offer limited gain.

B. Active RIS Model

In an active RIS system equipped with reflection type amplifiers, each element amplifies and adjust the phase shift of the incident signals. The i th element applies amplitude gain a_i and phase shift φ_i to the received signal. end-to-end signal-to-noise ratio (SNR) is

$$SNR_{active} = \frac{P_t \sum_{i=1}^N |a_i h_{BR,i} h_{RU,i} \exp(j\varphi_i)|^2}{\sum_{i=1}^N a_i^2 |h_{RU,i}|^2 \sigma^2 + \sigma^2} \quad (6)$$

where P_t is the transmit power, and σ^2 is the noise power.

The complex exponential $\exp(j\varphi_i)$ achieves phase alignment for constructive interference, while the amplitude factor a_i provides adaptive gain control. The absolute square operation converts the complex amplitude to signal power.

The denominator captures two different noise contributions:

- $\Sigma(a_i^2|h_{RU,i}|^2\sigma^2)$ represents the noise amplified by each active element.
- σ^2 considers the thermal noise of the UE receiver.

Equation (6) reveals the fundamental trade-off in active RIS: although amplification enhances signal strength, it also amplifies noise.

Total power consumption equation for active RIS

$$P_{total,active} = P_t + P_{amp} + P_{elements} + P_{control} + P_{proc} \quad (7)$$

which can be further expressed as

$$\begin{aligned} P_{total,active} = & 2P_t + N(P_{sw} + P_{dc})\eta_{sharing} \\ & + P_{static}(1 + 0.05\log_{10}N) \\ & + 0.001N\log_{10}(N + 1) \end{aligned} \quad (8)$$

where P_t is the transmit power, P_{amp} is the total amplification power which is equal to P_t , $P_{elements}$ is the total power consumption of the RIS elements, $P_{control}$ capturing logarithmic scaling, and P_{proc} is the computational processing power.

Since the total power consumption of RIS is the combination of P_{sw} and P_{dc} of all elements, so the $P_{elements} = N(P_{sw} + P_{dc})$. P_{sw} represents the power consumption of phase shifter circuits per element, and P_{dc} denotes the DC biasing power per amplifier element. Beside, $P_{control}$ is defined as $P_{control} = P_{sta}(1 + 0.05\log_{10}N)$ and $P_{proc} = 0.001N\log_{10}(N + 1)$. Meanwhile, $n_{sharing}$ represents the resource sharing advantage in large array, where $n_{sharing} = 1 - 0.1\log_{10}(N/100)$ and it is applied to the total power of RIS elements.

C. Passive RIS Model

For passive RIS, each element only adjusts the phase of the reflected signal without amplification. This phase distribution ensures that all signal components are constructively combined at the receiver, maximizing the received signal power. Therefore, the end-to-end SNR is defined as:

$$SNR_{pas} = \frac{P_t \sum_{i=1}^N (|h_{BR,i}| |h_{RU,i}| \eta_{phase} \eta_{coherent})^2}{\sigma^2} \quad (9)$$

where η_{phase} and $\eta_{coherent}$ are phase optimization and coherent merger efficiency.

Total power consumption equation for passive RIS

$$P_{total,passive} = P_t + P_{sw,total} + P_{control} \quad (10)$$

which can be further expressed as

$$P_{total,passive} = P_t + NP_{sw}\eta_{sharing} + P_{sta}(1 + 0.3\sqrt{\frac{N}{100}}) \quad (11)$$

where the total power consumption includes the transmit power P_t , Phase shifter power, $P_{sw,total} = NP_{sw}\eta_{sharing}$, and control power $P_{control} = P_{static}(1 + 0.3\sqrt{\frac{N}{100}})$. The definition of $\eta_{sharing}$ is the same as that of active RIS, reflecting the shared effect of control circuits in large arrays.

D. FAS Relay Model

The FAS relay uses N_{FAS} fluid antenna ports with adaptive port selection. Specifically, M optimal ports are selected based on the end to end channel gain. The number of selected ports follows an adaptive activation strategy: $M = \max(1, \text{round}(\text{selection}_{ratio} \times N_{FAS}))$, where the selection ratio adapts to array size: $\text{selection}_{ratio} = 0.15 - 0.05 \times \tanh((N_{FAS} - 100)/200)$. This adaptive approach provides approximately 15% activation for small arrays ($N < 100$) and gradually reduces to 10% for large arrays ($N > 500$), accounting for diminishing diversity returns due to spatial correlation in massive array adaptive port selection strategy, the FAS relay employs an adaptive port activation strategy rather than a fixed percentage. The activation ratio varies from 10% to 15% based on array size to optimize the diversity-correlation trade-off [10].

In AF protocol: the relay amplifies and forwards the received signal

$$SNR_{AF} = \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2 + 1} \eta_{hw} \quad (12)$$

In DF protocol: the relay decodes and re-encodes the signal

$$SNR_{DF} = \min(\gamma_1, \gamma_2) \eta_{hw} \quad (13)$$

where γ_1 means the signal to noise ratio experienced at the relay node for the base station to relay transmission and γ_2 is the signal to noise ratio experienced at the user equipment for the relay to user transmission; η_{hw} is the hardware implementation efficiency coefficient which is equal to 0.82.

The basic power consumption of the FAS relay system includes hardware power consumption components required by both the AF and DF protocols. This power consumption reflects the basic operating requirements of the FAS system and represents the electrical energy that must be consumed regardless of which relay protocol is used.

$$P_{base} = P_t + P_{relay} + P_{switches} + P_{amplifiers} + P_{control} + P_{sta} \quad (14)$$

where P_t is transmit power from base station; P_{relay} is relay transmission power; $P_{switches}$ is RF switch power $= 0.0008N$; $P_{amplifiers}$ is the amplifier power $= 0.007N_{selected}$, $P_{control}$ is the control circuit power $= 0.003N_{selected}$; P_{sta} is the static power consumption; $N_{selected}$ is the number of selected ports $= \max(1, \text{round}(\text{selection}_{ratio} \times N))$, selection_{ratio} is $0.15 - 0.05 \tanh((N - 100)/200)$.

Total power consumption equation for FAS relay

$$P_{total_{AF}} = P_{base} + P_{analog_{specific}} \quad (15)$$

which can be further expressed as

$$P_{total_{AF}} = P_t + P_{relay} + 0.0008N + 0.010N_{selected} + P_{sta} \quad (16)$$

where $P_{analog_{specific}} = 0.003 \times N_{selected}$.

$$P_{total_{DF}} = P_{base} + P_{digital_{specific}} + P_{codec_{specific}} \quad (17)$$

where $P_{digital_{specific}} = 0.005N_{selected}$; $P_{codec_{specific}} = 0.003N_{selected}$

$P_{total_{DF}}$ can be further expressed as

$$P_{total_{DF}} = P_t + P_{relay} + 0.0008N + 0.015N_{selected} + P_{sta} \quad (18)$$

where $N_{selected} = \max(1, \text{round}(0.15 - 0.05 \tanh(\frac{N-100}{200})))$.

III. OPTIMIZATION

A. RIS Optimization

For active RIS, the amplitude gain a_i is optimized under the total power constraint:

$$\max_{a_i, \varphi_i} SNR_{active} \quad (19)$$

$$s.t. \quad \Sigma(a_i^2(P_t h_{BR,i} + \sigma^2)) \leq P_{relay} \quad (20)$$

$$a_i \geq 0, \forall_i \in \{1, 2, \dots, N\} \quad (21)$$

$$\varphi_i \in (0, 2\pi), \forall_i \in \{1, 2, \dots, N\} \quad (22)$$

Using the Lagrange multiplier method, the optimal amplitude gain for the i th element is

$$a_i^* = \sqrt{P_{relay} / (N(P_t |h_{BR,i}| + \sigma^2))} \quad (23)$$

This solution ensures an even distribution of power among the elements while considering the channel conditions.

For passive RIS, the optimization focuses on phase alignment to obtain maximum coherent margin

$$\max_{\varphi_i} SNR_{passive} \quad (24)$$

$$s.t. \quad \varphi_i \in (0, 2\pi), \forall_i \in \{1, 2, \dots, N\} \quad (25)$$

The optimal phase of each element is

$$\varphi_i^* = \arg(h_{BR,i} * h_{RU,i}^*) \quad (26)$$

where $h_{BR,i}$ is the complex channel coefficient representing the magnitude and phase of the wireless link from the base station to the i th RIS element. $h_{RU,i}$ means from the i th RIS element to the user equipment. $h_{BR,i}^*$ is the complex conjugate of $h_{BR,i}$, $h_{RU,i}^*$ is the complex conjugate of $h_{RU,i}$

, they all obtained by negating the imaginary part, which reverses the phase while preserving the magnitude. $\arg(x)$ extracts the phase angle (in radians) of a complex number x .

This achieves perfect phase alignment and maximizes the coherent sum of the reflected signals.

B. FAS Relay Optimization

FAS relay optimization involves two key aspects:

The optimal M ports are selected based on the end-to-end channel gain

$$S = \arg \max_{|S|=M} \sum_{i \in S} |g_{BR,i}| |g_{RU,i}| \quad (27)$$

where $g_{BR,i}$ and $g_{RU,i}$ are the base station to i th port and port to user equipment channel gains, respectively.

To optimize the inter-hop power distribution. Golden section search is used. The problem can be formulated as

$$\min_{P_1, P_2} P_1 + P_2 \quad (28)$$

$$s.t. \quad C_{total} \geq C_{min} \quad (29)$$

$$P_1 + P_2 \leq P_{total} \quad (30)$$

where the golden section method optimizes the power allocation iteratively with a convergence tolerance $\varepsilon = 10^{-8}$, P_1 is first hop power (base station to FAS relay), P_2 is second hop power (FAS relay to user)

For all systems, the rate is defined as

$$C_{total} = \log_2(1 + SNR) \quad (31)$$

and the energy efficiency (EE) is defined as

$$EE = C_{total} / P_{total} \quad (32)$$

The optimization balances spectral efficiency gains with increased power consumption, which is particularly important for large-scale deployments where power consumption scales with array size. Therefore, the optimization framework looks at the trade-off between power use and array gain.

IV. RESULTS

This section presents the monte carlo simulation results comparing Active RIS, Passive RIS, and FAS relay systems under different operational scenarios. With a $1.0\text{m} \times 1.0\text{m}$ square panel and 2.5 GHz carrier frequency, the system has 256 RIS elements and 1089 FAS ports, with spacing requirements of $\lambda/2$ for RIS and $\lambda/4$ for FAS. The channel model uses Rician fading (K -factor = 10), and $\alpha = 2.2$. $\rho = 0.7$ for RIS elements, $\rho = 0.8$ for FAS ports, with $\rho = 0$ used as the independent fading baseline for comparison. System distances are $d_{BR} = d_{RU} = 400\text{m}$.

The performance ranking from highest to lowest is as follows: Active RIS achieves 7.1 bps/Hz, followed by FAS Relay DF with 7.6 bps/Hz, then FAS Relay AF with 4.7 bps/Hz, and finally Passive RIS with 3.8 bps/Hz. Besides, the weaker hops of the dual-hop system dictate the end to

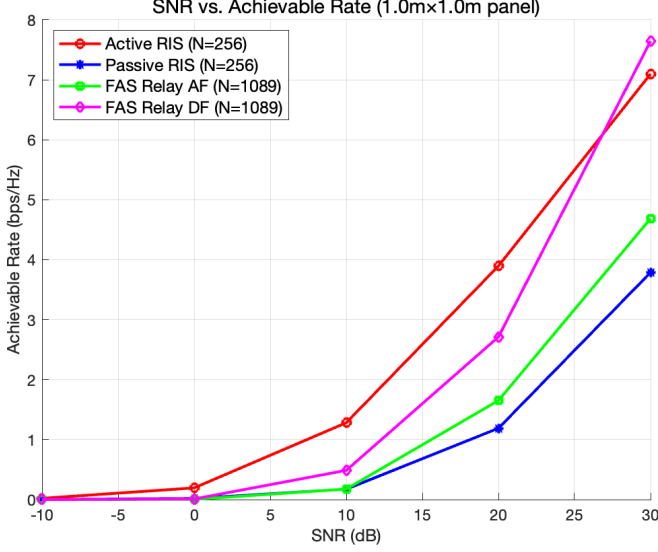


Fig. 1. Rate versus SNR

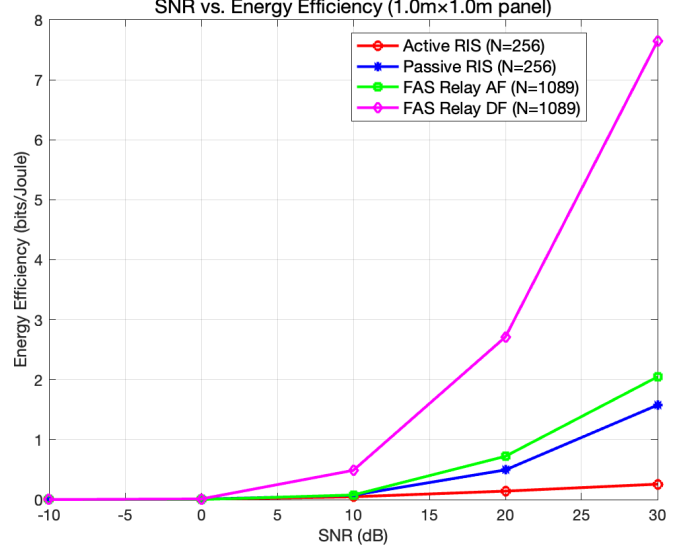


Fig. 2. Energy Efficiency versus SNR

end performance of the DF FAS relay, limiting the benefits of stronger hops. To overcome this limitation, the port selection method optimizes bases on (26), picking 154 ideal ports (15% activation rate). The DF protocol grows rapidly in high SNR regions and approaches active RIS performance. In addition, FAS relay AF strikes a balance between two hops, albeit at the expense of noise amplification, unlike DF's minimum rate constraint. Each engaged port increases received noise, rising total noise power which affect low-to-medium SNR performance. Passive RIS performs worst as signal phase alignment is only possible with $\varphi_i^* = -\angle(h_{BR,i}) - \angle(h_{RU,i})$ and cannot overcome power attenuation in dual-hop systems.

In Figure 2, the combination of high rate performance and low power consumption makes DF FAS relay the most energy efficient compared to other technologies. The DF protocol optimizes signal quality by preventing noise amplification, while selective port activation reduces power consumption by 85% compared to full-port activation. Adaptive port selection maximizes energy savings by using only the most efficient ports. Active RIS has the fastest data rate, but its power consumption increases faster than performance gains, resulting in low energy efficiency. Through selective activation, AF FAS relay outperforms passive RIS in terms of energy efficiency since the performance of passive RIS is constraint by the double fading issue.

Figures 3 show how array dimension scaling affects rate performance. Note that the x axis shows the number of elements L per row or column, where $L \times L = N$. Active RIS shows optimal linear growth, increasing from $0.25\text{bps}/\text{Hz}$ with 2×2 elements to $2.7\text{bps}/\text{Hz}$ with 10×10 elements, following the array gain scaling law. In contrast, FAS technology and passive RIS exhibit performance saturation in large arrays. The slow growth of DF FAS relay is primarily caused by the

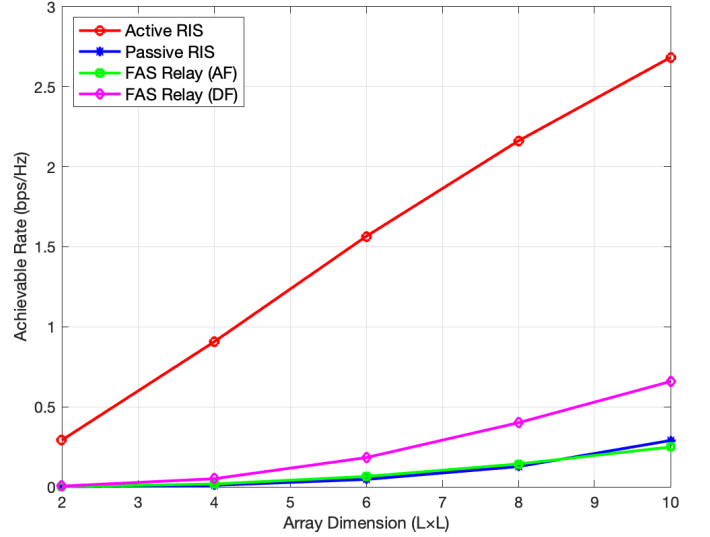


Fig. 3. Rate versus Array Dimension

diminishing marginal benefits of the port selection algorithm: once the number of ports reaches a certain scale, adding new ports is unlikely to provide significant improvements in end-to-end channel gain. The saturation effect of passive RIS stems from the influence of spatial correlation. As the array size increases, the channel correlation between adjacent elements limits further improvements in diversity gain.

Figure 4 shows the scaling dynamic for energy efficiency. Active RIS has the highest energy efficiency, at 2×2 arrays where performance-power trade-off reaches optimal balance. However, when the array dimension increases, the energy effi-

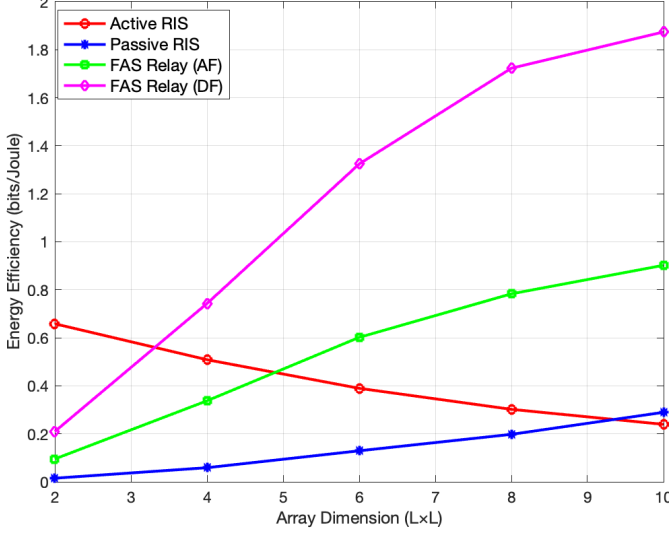


Fig. 4. Energy Efficiency versus Array Dimension

ciency decreases as mandatory full-element activation causes linear power scaling to outpace logarithmic performance improvements. The law of declining marginal returns applies when power consumption growth surpasses performance gains. On the other hand, the energy efficiency of FAS relay DF is the best overall. As the array size increases, the port selection method becomes more efficient, allowing the fixed activation ratio to select higher-quality port combinations and enhance DF FAS relay performance.

In summary, although active RIS has the best rate performance, it suffers from poor energy efficiency performance due to the power consumption of the amplifiers which scales with the number of active elements. FAS relay DF has the best energy efficiency performance overall and the rate performance is comparable to active RIS at high SNR and significantly better than FAS relay AF and passive RIS in the whole range of SNR.

V. CONCLUSION

This paper aim to compare active RIS, passive RIS, and FAS relay performance. The results show that even though active RIS has the best rate performance, it suffers from poor energy efficiency performance due to the power consumption of the amplifiers which scales with the number of active elements. FAS relay DF has the best energy efficiency performance overall and the rate performance is comparable to active RIS at high SNR and significantly better than FAS relay AF and passive RIS in the whole range of SNR. FAS technology's energy efficiency benefits from the increase of arrange dimension.

ACKNOWLEDGMENT

This work was supported by the Ministry of Higher Education Malaysia through the Higher Institution Centre of Excellence (HICOE) Grant 4J636 and in part by EU HORIZON

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